The Chi-Square Distribution

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Question 1

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Part 1 of 5

The numbers of false fire alarms were counted each month at a number of sites. The results are given in the following table.

Month	Number of Alarms
January	36
February	40
March	42
April	25
May	40
June	45
July	38
August	42
September	26
October	27
November	25
December	42

Test the hypothesis that false alarms are equally likely to occur in any month. Use 1% level of significance.

Procedure: Select an answer

Assumptions: (select everything that applies)

- ☐ Normal populations
- ☐ Equal population standard deviations
- ☐ At most 20% of the expected frequencies are less than 5
- ☐ Independent samples
- ☐ All expected frequencies are 1 or greater
- ☐ Simple random sample

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Question 2

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Part 1 of 5

The table below shows the distribution of ER visits across the days of the week in a random sample of 390 visits at a random local hospital in a large city:

Days	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Visits	43	67	75	66	51	42	46

Are ER visits evenly distributed across the days of the week? Use 10% level of significance.

Procedure: Select an answer

Assumptions: (select everything that applies)

- At most 20% of the expected frequencies are less than 5
- All expected frequencies are 1 or greater
- Equal population standard deviations
- Independent samples
- Simple random sample
- Normal populations

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Question 3

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You are conducting a test of homogeneity for the claim that two different populations have the same proportions of the following two characteristics. Here is the sample data.

Category	Population #1	Population #2
A	13	97
B	51	57

The expected observations for this table would be

Category	Population #1	Population #2
A		
B		

The resulting squared Pearson residuals are:

Category	Population #1	Population #2
A		
B		

What is the chi-square test-statistic for this data?

$\chi^2 =$ ______

Report all answers accurate to three decimal places. (Though you may need to retain more decimal places in the residuals to get the correct chi-square test statistic.)

Hint 1

Question 4 0/1 pt 3 19

The 2010 general Social Survey asked a large number of people how much time they spent watching TV each day. The mean number of hours was 4.41 with a standard deviation of 2.01. Assume that in a sample of 30 teenagers, the sample standard deviation of daily TV time is 2.04 hours, and that the population of TV watching times is normally distributed. Under 10% significance level can you conclude that the population standard deviation of TV watching times for teenagers is different from 2.01?

Procedure: Select an answer

Assumptions: (select everything that applies)

- The number of positive and negative responses are both greater than 10
- Sample size is greater than 30
- Population standard deviation is unknown
- Normal population
- Simple random sample
- Population standard deviation is known

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